

📘 AI/ML Internship – Day 30 Notes & Tasks

🟢 Module 6: Data Visualization Basics with Matplotlib

📌 Part 1: What is Data Visualization?

💠 Definition

👉 Data visualization means representing data using:

- Charts
- Graphs
- Plots

👉 Instead of reading large tables, graphs make data easier to understand.

📌 Part 2: Why Visualization is Important?

👉 In AI/ML:

- Helps identify patterns
- Detect trends
- Understand datasets visually

👉 Example:

- Sales growth
 - Student performance
 - Temperature changes
-

📌 Part 3: Theory Answers (IMPORTANT)

💠 1. What is Matplotlib?

👉 Matplotlib is a Python library used for creating graphs and charts.

◆ 2. Why Visualization is Important?

👉 Visualization:

- Makes data easy to understand
 - Helps analyze patterns
 - Improves decision making
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◆ 3. What is a Plot?

👉 A plot is a graphical representation of data.

◆ 4. What is X-axis and Y-axis?

Axis Meaning

X-axis Horizontal axis

Y-axis Vertical axis

◆ 5. Why Graphs are Used in AI?

👉 Graphs help visualize:

- Trends
 - Accuracy
 - Predictions
 - Model performance
-

📌 Part 4: Installing Matplotlib

`pip install matplotlib`

📌 Part 5: Import Matplotlib

`import matplotlib.pyplot as plt`

✦ Part 6: Simple Line Plot

```
import matplotlib.pyplot as plt
```

```
x = [1, 2, 3, 4]
```

```
y = [10, 20, 30, 40]
```

```
plt.plot(x, y)
```

```
plt.show()
```

✦ Part 7: Adding Labels & Title

```
plt.plot(x, y)
```

```
plt.title("Student Marks")
```

```
plt.xlabel("Subjects")
```

```
plt.ylabel("Marks")
```

```
plt.show()
```

✦ Part 8: Changing Line Style

```
plt.plot(x, y, linestyle="--", marker="o")
```

```
plt.show()
```

✦ Part 9: Bar Chart

- ◆ Used for Comparison

```
students = ["A", "B", "C"]
```

```
marks = [80, 90, 75]
```

```
plt.bar(students, marks)
```

```
plt.show()
```

📌 Part 10: Pie Chart

- ◆ Used for Percentage Representation

```
data = [40, 30, 20, 10]
```

```
labels = ["AI", "Web", "Cloud", "Cyber"]
```

```
plt.pie(data, labels=labels)
```

```
plt.show()
```

📌 Part 11: Scatter Plot

- ◆ Used for Relationship Analysis

```
x = [1, 2, 3, 4]
```

```
y = [10, 15, 20, 25]
```

```
plt.scatter(x, y)
```

```
plt.show()
```

📌 Part 12: Grid & Legend

```
plt.plot(x, y, label="Marks")
```

```
plt.legend()
```

```
plt.grid()
```

```
plt.show()
```

📌 Part 13: Real-World Thinking

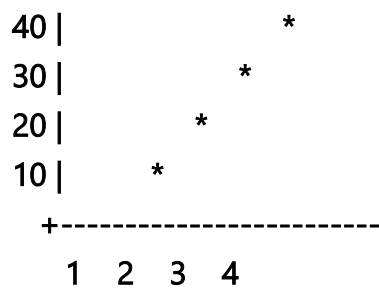
👉 In AI/Data Science:

- Accuracy graphs
- Loss graphs
- Sales analysis
- Weather visualization

👉 Visualization helps explain results clearly.

📌 Part 14: Visualization Example

Marks Graph



🎯 Tasks for Day 30

🧠 Task 1: Theory (Write Answers)

1. What is Matplotlib?
 2. Why visualization is important?
 3. What is plot?
 4. Difference between X-axis and Y-axis
 5. Why graphs are important in AI?
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💻 Task 2: Line Plot Practice

1. Create simple line graph

2. Add labels
 3. Add title
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Task 3: Styling Graphs

1. Add marker
 2. Change line style
 3. Add grid
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Task 4: Bar Chart Practice

1. Student marks comparison
 2. Sales comparison
 3. Product comparison
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Task 5: Pie Chart Practice

1. Department distribution
 2. Expense analysis
 3. Course popularity
-

Task 6: Scatter Plot Practice

1. Height vs weight
2. Study hours vs marks
3. Temperature vs sales